NATIONAL UNIVERSITY OF SINGAPORE
Department of Statistics and Data Science

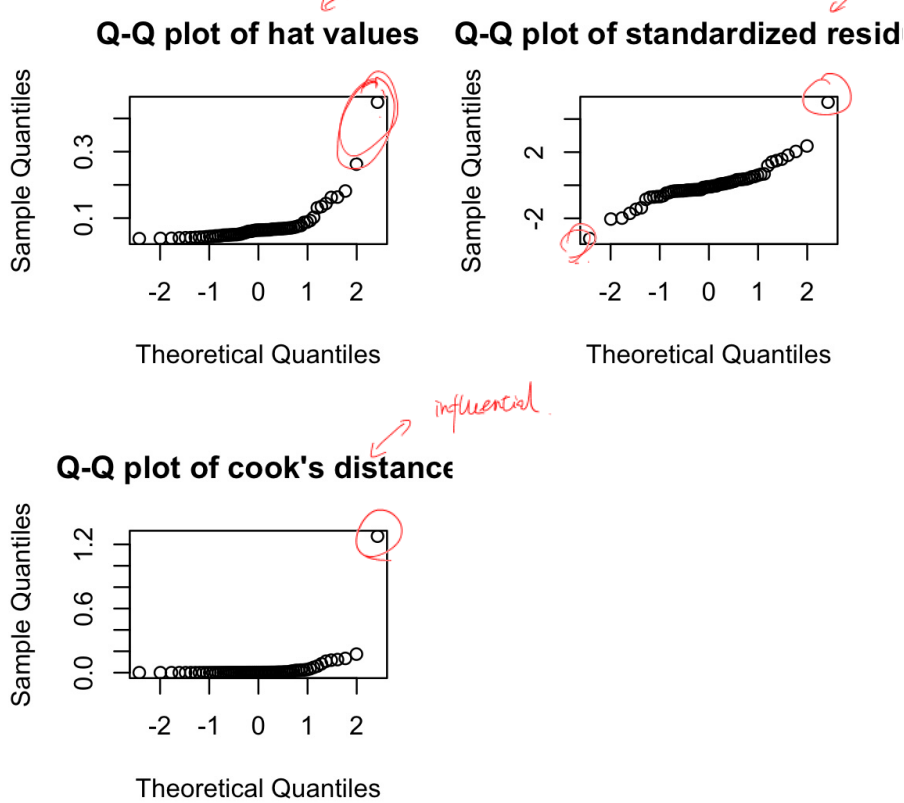
ST5202: Applied Regression Analysis

TUTORIAL 10

1. For the **Employee salary (ES) data** dealt with in Lecture notes 8, consider `current salary` as the response variable and the predictor variables `degree obtained`, `years` of experience since last degree and `number of persons currently supervised`.

   (i) Fit the unweighted regression model. The preliminary analysis also reveals that there are influential outliers. Remove the observations with largest leverage, cook's distance, and studentized deleted residual (in absolute value).

leverage



influencial

From the preliminary analysis, the points with largest statistics are as follows

Hat Value.

```
55    55 1.222318e-01    38 0.16328962    55   1.80983100
65    65 1.332828e-01    11 0.18156010    61   2.04121127
23    23 1.734826e-01    43 0.26240556    19   2.37350001
43    43 1.276863e+00     3 0.44890756    43   5.01764585
```

Hence, observation 3 has the largest leverage, while observation 43 has both the largest Cook's distance and the largest studentized deleted residual in absolute value. We will remove 3 and 43.

(ii) After removing them, refit the unweighted regression model. By using the residual plots against fitted values, check whether or not there is evidence of non-constant variance.

After removing observations 3 and 43, the unweighted regression model is refitted. The fitting result is as follows.

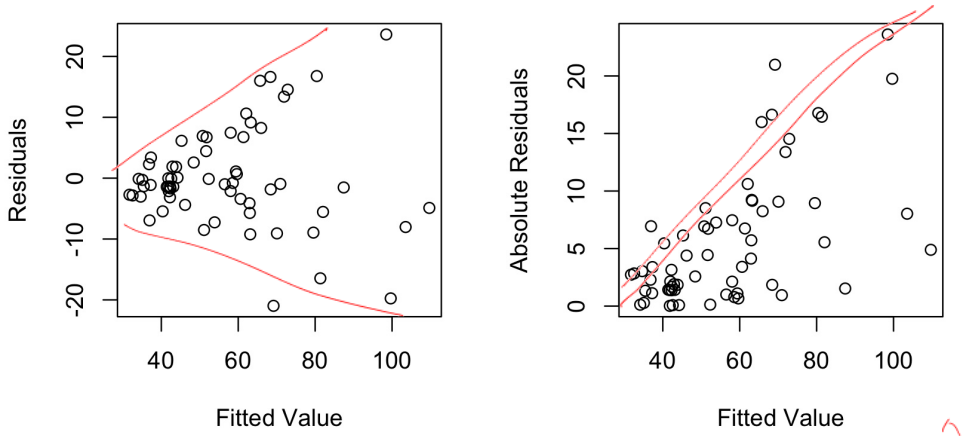
```
> summary(uw.fit)

Call:
glm(formula = y ~ D + x1 + x2, data = wk.dat3)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  30.7661     2.3892  12.877  < 2e-16 ***
D2           10.0072     2.6886   3.722 0.000448 ***
D3           25.5311     2.9608   8.623 5.58e-12 ***
x1            1.4771     0.1945   7.595 2.93e-10 ***
x2            1.0055     0.2959   3.398 0.001233 **
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

    Null deviance: 25517.4  on 62  degrees of freedom
Residual deviance:  4126.9  on 58  degrees of freedom
```

The residual plot against fitted values still shows that the spread of the residuals increases as the fitted values increase. The plot of absolute residuals against fitted values also shows an increasing trend. Hence there is still evidence of non-constant variance.



$\hat{r}_i \to w = 1/\hat{r}_i^2$

(iii) Estimate a suitable set of weights by regressing $\log|r_i|$ on $\log \hat{y}_i$, where r_i and $\hat{y}_i$ are respectively the residual and fitted value from the refitted unweighted model in (ii). Fit the corresponding weighted regression model, and check by using the appropriate weighted residual plot whether the non-constant variance problem has been rectified.

Fit the regression model

$$\log|r_i| = \alpha_0 + \alpha_1 \log \hat{y}_i + e_i.$$

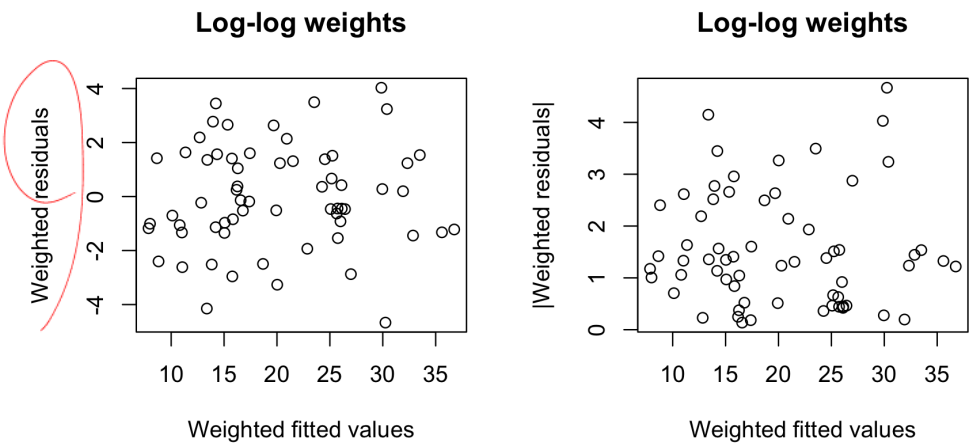
The weight is estimated as $w_i = 1/\hat{r}_i^2$.

Use these estimated weights, the resulting weighted regression fit is

$$\hat{y} = 28.9863 + 11.0468D_2 + 26.8896D_3 + 1.5522x_1 + 1.5244x_2.$$

All the regression coefficients are highly significant. The weighted residual plot no longer shows the increasing spread seen in the unweighted fit. Hence the non-constant variance problem has been substantially rectified.

$W^{1/2} Y = W^{1/2} X\beta + W^{1/2}\varepsilon$



(iv) Estimate another set of weights by regressing $|r_i|$ on $\hat{y}_i$, where r_i and $\hat{y}_i$ are respectively the residual and fitted value from the refitted unweighted model in (ii). Fit the corresponding weighted regression model, and check by using the appropriate weighted residual plot whether the non-constant variance problem has been rectified.

Using the regression

$$|r_i| = \beta_0 + \beta_1 \hat{y}_i + e_i,$$

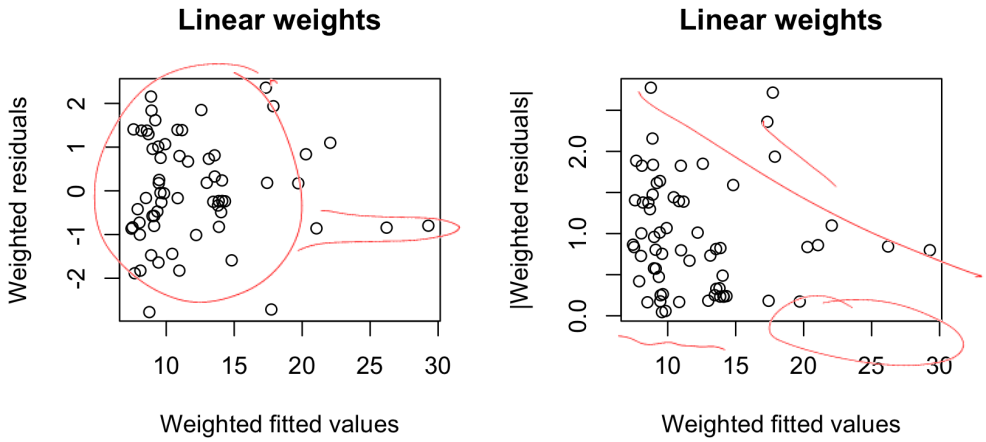
$\hat{r}_i$ $\quad w_i = \frac{1}{\hat{r}_i^2}$

and the weights are taken as $w_i = \frac{1}{\hat{s}_i^2}$.

The resulting weighted regression fit is

$$\hat{y} = 28.7789 + 11.2001 D_2 + 26.8285 D_3 + 1.5899 x_1 + 1.2831 x_2.$$

Again, all the regression coefficients are significant. However, the weighted residual plot still shows a non-constant variance pattern, in which the standard deviation becomes smaller when the fitted values are large. Hence this choice of weights does not rectify the non-constant variance problem satisfactorily.



(v) Briefly compare the weighted fits in (iii) and (iv).

The log-log weighting scheme in (iii) performs better than the linear weighting scheme in (iv). Although the two weighted fits give similar coefficient estimates, only the log-log weighted residual plot does not show obvious evidence of non-constant variance. In contrast, for the linear weighting scheme, the weighted residual plot still shows a variance pattern, with smaller spread for larger fitted values. Therefore the log-log weighting scheme should be preferred.

2. Consider the linear regression model

$$y = X\beta + \varepsilon,$$

where

$$\varepsilon \sim N(0, \Sigma), \qquad \Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2) = \sigma^2 W^{-1}.$$

(i) Write down the log-likelihood function of the model parameters, up to an additive constant.

Since $y \sim N(X\beta, \Sigma)$, the log-likelihood is $\sum_{i=1}^{n} \frac{(y_i - x_i^T\beta)^2}{\sigma_i^2}$ $\sum_{i=1}^{n} \log \sigma_i^2$

$y \sim N(X\beta, \Sigma)$

$f(\beta, \Sigma) = \frac{1}{(2\pi)^{n/2}} \cdot \frac{1}{\sqrt{|\Sigma|}} \cdot \exp\{-\frac{1}{2}(y - X\beta)^T \Sigma^{-1} (y - X\beta)\}$

$$\ell(\beta, \sigma^2) = -\frac{1}{2}\log|\Sigma| - \frac{1}{2}(y - X\beta)^T \Sigma^{-1}(y - X\beta) + C,$$

where C is a constant free of the parameters. Since $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$, this can be written as

$$\ell(\beta, \sigma^2) = -\frac{1}{2}\sum_{i=1}^{n} \log(\sigma_i^2) - \frac{1}{2}\sum_{i=1}^{n} \frac{(y_i - x_i^T\beta)^2}{\sigma_i^2} + C.$$

Equivalently, using $\Sigma = \sigma^2 W^{-1}$,

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log\sigma^2 + \frac{1}{2}\log|W| - \frac{1}{2\sigma^2}(y - X\beta)^T W(y - X\beta) + C.$$

(ii) Find the maximum likelihood estimator of $\boldsymbol{\beta}$.

For fixed σ^2*, maximizing the log-likelihood is equivalent to minimizing*

$$Q(\beta) = (y - X\beta)^T W(y - X\beta).$$

Differentiate with respect to β*:*

$$\frac{\partial Q(\beta)}{\partial \beta} = -2X^T W(y - X\beta).$$

Set the derivative to zero:

$$X^T W X\,\hat{\beta} = X^T W y.$$

Hence the maximum likelihood estimator of β *is*

$$\hat{\beta} = (X^T W X)^{-1} X^T W y.$$

WLSE:

$W^{1/2} y = W^{1/2} X\beta + W^{1/2}\varepsilon$

$\hat{\beta} = (X^T W X)^{-1} X^T W y$

This is the weighted least squares estimator.

3. Consider the data set `longley` in R. The data can be loaded by using the command line

```
data(longley)
```

Regress `Employed` on the predictor variables `GNP.deflator`, `GNP`, `Unemployed`, `Armed.Forces`, `Population` and `Year`. In this course, variance inflation factor values greater than 5 are taken as indicating considerable multicollinearity.

(i) Fit the full regression model and compute the variance inflation factors. Identify the predictor variable for which the multicollinearity appears to be most serious.

The variance inflation factors of the full model are

```
GNP.deflator          GNP   Unemployed Armed.Forces   Population         Year
   135.53244   1788.51348     33.61889      3.58893    399.15102    758.98060
```

Since several VIF values are much greater than 5, the multicollinearity is very serious. The largest VIF is attained by ***GNP****, with value 1788.51348. Hence* ***GNP*** *is the predictor variable for which the multicollinearity appears to be most serious.*

(ii) Remove the predictor variable identified in (i). Fit the resulting reduced model and compute the variance inflation factors again. Identify the predictor variable for which the multicollinearity now appears to be most serious.

After removing ***GNP****, the reduced model is fitted. The corresponding output is*

```
Call:
lm(formula = Employed ~ GNP.deflator + Unemployed + Armed.Forces +
    Population + Year, data = longley, x = TRUE)

Residuals:
     Min       1Q   Median       3Q      Max
-0.47664 -0.12157 -0.01118  0.10669  0.50613

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  -2.705e+03  5.182e+02  -5.220 0.000390 ***
GNP.deflator -4.392e-02  6.503e-02  -0.675 0.514806
Unemployed   -1.526e-02  1.600e-03  -9.538 2.45e-06 ***
Armed.Forces -9.258e-03  1.906e-03  -4.856 0.000665 ***
Population   -2.526e-01  1.259e-01  -2.006 0.072661 .
Year          1.439e+00  2.742e-01   5.246 0.000375 ***
---
Residual standard error: 0.307 on 10 degrees of freedom
Multiple R-squared:  0.9949,    Adjusted R-squared:  0.9924
F-statistic: 390.5 on 5 and 10 DF,  p-value: 4.004e-11
```

The variance inflation factors of this reduced model are

```
GNP.deflator   Unemployed Armed.Forces   Population         Year
   78.372376     3.557776     2.800835   122.047678   271.189670
```

Thus the largest VIF is now attained by ***Year****, with value 271.189670. Hence* ***Year*** *is now the predictor variable for which the multicollinearity appears to be most serious.*

(iii) Remove the predictor variable identified in (ii). Fit the resulting reduced model and compute the variance inflation factors again. Briefly comment on whether the multicollinearity problem has been substantially reduced.

After removing ***Year*** *as well, the further reduced model is fitted. The corresponding output is*

```
Call:
lm(formula = Employed ~ GNP.deflator + Unemployed + Armed.Forces +
```

```
    Population, data = longley, x = TRUE)

Residuals:
     Min       1Q   Median       3Q      Max
-0.65617 -0.25766 -0.00081  0.12135  1.22254

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  13.781314   6.886470   2.001 0.070657 .
GNP.deflator  0.207046   0.081376   2.544 0.027270 *
Unemployed   -0.012412   0.002780  -4.465 0.000955 ***
Armed.Forces -0.005968   0.003325  -1.795 0.100170
Population    0.306601   0.123795   2.477 0.030755 *
---
Residual standard error: 0.5671 on 11 degrees of freedom
Multiple R-squared:  0.9809,    Adjusted R-squared:  0.9739
F-statistic: 141.1 on 4 and 11 DF,  p-value: 2.26e-09
```

The variance inflation factors are now

```
GNP.deflator   Unemployed Armed.Forces   Population
   35.970754     3.147600     2.497795    34.588299
```

The multicollinearity problem has been reduced, since the VIF values are much smaller than those in the full model. However, it has not been substantially rectified, because the VIF values of `GNP.deflator` *and* `Population` *are still much larger than 5. Hence considerable multicollinearity still remains in the model.*

4. The data set `mtcars` consists of the following variables: `mpg`, `cyl`, `disp`, `hp`, `drat`, `wt`, `qsec`, `vs`, `am`, `gear`, `carb`.

   (i) Consider the regression model with `mpg` as the response variable and all the other variables as the predictor variables. Fit the full regression model and compute the variance inflation factors. Identify the predictor variables for which the multicollinearity appears to be serious.

   The output of the full regression model is

```
Call:
lm(formula = mpg ~ cyl + disp + hp + drat + wt + qsec + vs +
    am + gear + carb, data = mtcars, x = TRUE)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.30337   18.71788   0.657   0.5181
cyl         -0.11144    1.04502  -0.107   0.9161
disp         0.01334    0.01786   0.747   0.4635
hp          -0.02148    0.02177  -0.987   0.3350
drat         0.78711    1.63537   0.481   0.6353
wt          -3.71530    1.89441  -1.961   0.0633 .
qsec         0.82104    0.73084   1.123   0.2739
vs           0.31776    2.10451   0.151   0.8814
am           2.52023    2.05665   1.225   0.2340
gear         0.65541    1.49326   0.439   0.6652
carb        -0.19942    0.82875  -0.241   0.8122
```

The corresponding variance inflation factors are

```
       cyl      disp        hp      drat        wt      qsec        vs        am
 15.373833 21.620241  9.832037  3.374620 15.164887  7.527958  4.965873  4.648487
      gear      carb
  5.357452  7.908747
```

Since values of VIF greater than 5 indicate considerable multicollinearity, the multicollinearity appears to be serious for `disp`, `cyl`, `wt`, `hp`, `qsec`, `gear` *and* `carb`. *In particular,* `disp` *has the largest VIF, followed by* `cyl` *and* `wt`. *Hence the full model suffers from serious multicollinearity.*

(ii) Using ridge regression to fit the model in the presence of multicollinearity. Select the tuning parameter by using an appropriate cross-validation criterion.

Using ridge regression and selecting the tuning parameter by the generalized cross-validation criterion, the relevant output is

```
> regRidge.best
                      cyl          disp            hp          drat            wt
21.135473178 -0.373031194 -0.005272819 -0.011598352  1.053480536 -1.230197747
        qsec            vs            am          gear          carb
 0.161653471  0.771183921  1.619830781  0.544234063 -0.546670017

> regRidge.best$lambda
[1] 14.92

> regRidge.best$GCV
0.2176177
```

Hence the tuning parameter selected by GCV is

$$\lambda = 14.92.$$

This value gives the minimum GCV score 0.2176177, and the corresponding ridge regression fit is taken as the final ridge fit.

(iii) Using principal component regression to fit the model. Determine the number of principal components by using an appropriate criterion.

Using principal component regression with cross-validation, the relevant output is

```
> summary(pcr.fit)
Data:  X dimension: 32 10
       Y dimension: 32 1
Fit method: svdpc
Number of components considered: 10

VALIDATION: RMSEP
Cross-validated using 10 random segments.
       (Intercept)  1 comps  2 comps  3 comps  4 comps  5 comps  6 comps
CV           6.123    2.589    2.609    2.519    2.536    2.571    2.679
adjCV        6.123    2.580    2.600    2.505    2.521    2.555    2.657
       7 comps  8 comps  9 comps  10 comps
CV       2.738    2.941    3.554     3.448
adjCV    2.712    2.898    3.469     3.364

TRAINING: % variance explained
     1 comps  2 comps  3 comps  4 comps  5 comps  6 comps  7 comps  8 comps
X      57.60    84.10    90.07    92.77    94.99    97.09    98.42    99.23
mpg    82.53    82.63    85.40    85.41    85.47    85.56    85.58    85.85
     9 comps  10 comps
X      99.76     100.0
mpg    86.09      86.9

> ncomp.best
3 comps
      3

> summary(pcr.best)
Data:  X dimension: 32 10
       Y dimension: 32 1
```

$X^T X$

$\frac{\sum_{i=1}^{3} \lambda_i}{\sum_{i=1}^{10} \lambda_i} = Tr(X^T X)$ $\lambda_i (X^T X)$

```
Fit method: svdpc
Number of components considered: 3
TRAINING: % variance explained
     1 comps  2 comps  3 comps
X      57.60    84.10    90.07
mpg    82.53    82.63    85.40
```

The cross-validated RMSEP is smallest at 3 principal components. Hence the appropriate number of principal components is 3. With 3 principal components, about 90.07% of the variation in the predictors and 85.40% of the variation in `mpg` *are explained.*

(iv) Compare the ridge regression fit and the principal component regression fit for prediction purpose. Briefly comment on the use of these two methods as remedies for multicollinearity.

The mean squared prediction errors are

$$MSE_{\text{ridge}} = 5.385876, \qquad MSE_{\text{PCR}} = 5.136009.$$

Hence the principal component regression fit performs slightly better than the ridge regression fit for prediction in this example.

Both ridge regression and principal component regression are suitable remedies for multicollinearity when the main purpose is prediction. Ridge regression uses all the predictors but shrinks the regression coefficients toward zero, thereby stabilizing the estimation. Principal component regression replaces the original predictors by a smaller number of principal components, and thus reduces the effect of strong linear dependence among the predictors.

Therefore both methods are appropriate for handling serious multicollinearity. However, they are mainly used for building prediction models, rather than for direct interpretation of the individual regression coefficients.